Chapter 1

Linear Systems

1.1 Introduction

Solving systems of linear equations is one of the most important applications of linear algebra, if not the most important. Linear systems play a pivotal role in many areas of mathematics, including the following:

- Numerical solution of PDEs/ODEs
- Optimization
- Nonlinear equation solving
- Regression and least-squares fitting

These are just a few examples.

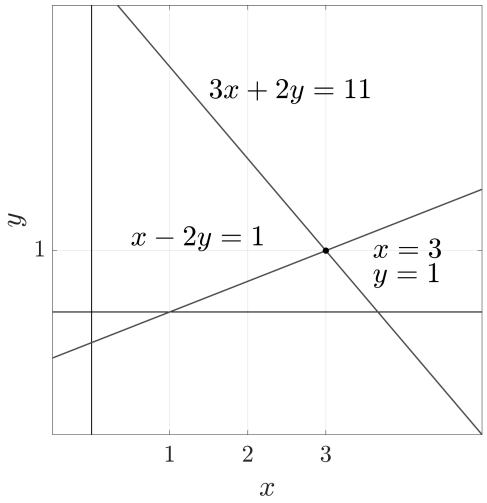
1.2 Ways of viewing a linear system

Let us start with a simple 2×2 linear system:

$$x - 2y = 1 \tag{1.1}$$

$$3x + 2y = 11 \tag{1.2}$$

We can look at this system in different ways. We begin by considering **one row at a time**. The two equations represent two straight lines. We can plot them in the xy-plane:

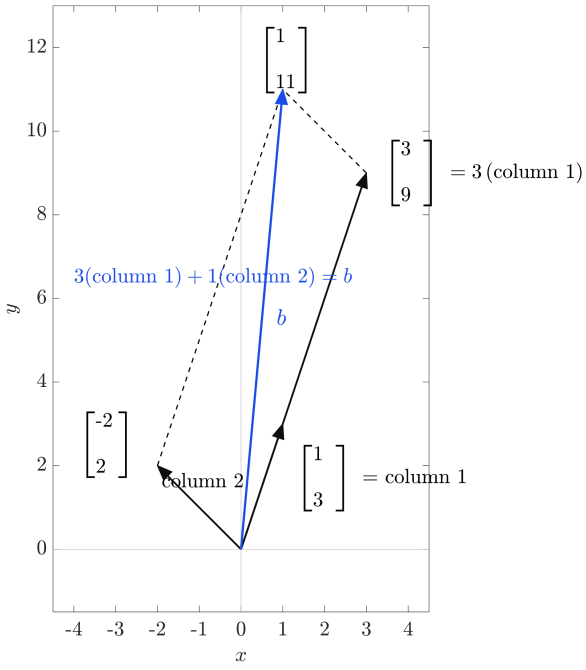


All the points on the $x - 2y = 1$ line satisfy the first equation, and all the points on the $3x + 2y = 11$ line satisfy the second. But only one point satisfies both: the intersection at $(x, y)^T = (3, 1)^T$. This is the familiar "**row picture**".

Turn now to the column picture. We can write the system as follows:

$$x \begin{pmatrix} 1 \\ 3 \end{pmatrix} + y \begin{pmatrix} -2 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ 11 \end{pmatrix} = \mathbf{b} \tag{1.3}$$

This has two column vectors on the left side. The goal is to find the combination of those vectors that equals the vector on the right. We are multiplying the first column by x and the second column by y, and then adding the results. With the choices $x = 3$ and $y = 1$ (the same values as before), this gives three times the first column plus the second column, which equals $\mathbf{b}$.



The third way of looking at the system is to use matrices. The matrix form is

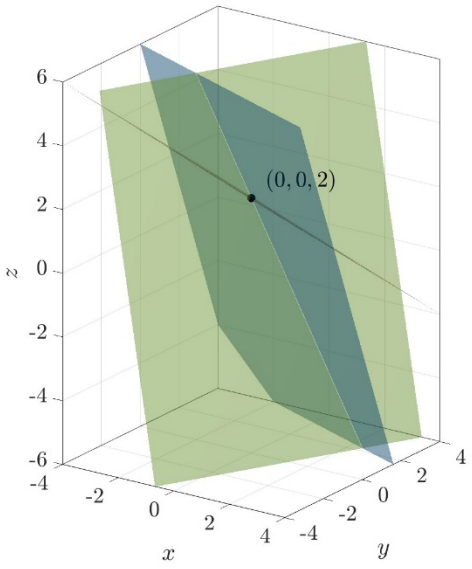
$$\begin{pmatrix} 1 & -2 \\ 3 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 11 \end{pmatrix} \tag{1.4}$$

The multiplication $A\mathbf{x}$ produces a vector. We seek a vector $\mathbf{x} = (x, y)^T$ whose product with A, namely $A\mathbf{x}$, equals the given vector $\mathbf{b}$.

Example: Now consider a 3×3 system:

$$\begin{aligned} x + 2y + 3z &= 6 \\ 2x + 5y + 2z &= 4 \\ 6x - 3y + z &= 2 \end{aligned} \tag{1.5}$$

In the row picture, the three equations represent three planes. The first two planes intersect along a line. Every point on this line satisfies the first two equations. The third plane intersects this line at a single point. This point satisfies all three equations and is therefore the desired solution.



What if we have a 10×10 system? The row picture would involve ten hyperplanes, each a nine-dimensional plane in ten-dimensional space. Their common intersection would give the solution set. Such a picture is difficult to visualize, so let us return to the three-dimensional example and consider its column picture:

$$x \begin{pmatrix} 1 \\ 2 \\ 6 \end{pmatrix} + y \begin{pmatrix} 2 \\ 5 \\ -3 \end{pmatrix} + z \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 6 \\ 4 \\ 2 \end{pmatrix} = \mathbf{b} \tag{1.6}$$

The required combination is:

$$0\begin{pmatrix}1\\2\\6\end{pmatrix}+0\begin{pmatrix}2\\5\\-3\end{pmatrix}+2\begin{pmatrix}3\\2\\1\end{pmatrix}=\begin{pmatrix}6\\4\\2\end{pmatrix}=\mathbf{b} \tag{1.7}$$

Now consider the matrix picture:

$$\begin{pmatrix}1&2&3\\2&5&2\\6&-3&1\end{pmatrix}\begin{pmatrix}x\\y\\z\end{pmatrix}=\begin{pmatrix}6\\4\\2\end{pmatrix} \tag{1.8}$$

A basic question is: What does it mean to multiply A by $\mathbf{x}$? We can multiply by rows or by columns. If we multiply by rows, then we are essentially using the concept of the dot product:

$$A\mathbf{x}=\begin{pmatrix}A_{1*}\cdot\mathbf{x}\\A_{2*}\cdot\mathbf{x}\\A_{3*}\cdot\mathbf{x}\end{pmatrix} \tag{1.9}$$

However, if we multiply by columns,

$$A\mathbf{x}=x_1A_{*1}+x_2A_{*2}+x_3A_{*3} \tag{1.10}$$

□

1.3 Gaussian Elimination—Regular Case

A standard technique for solving a linear system is **Gaussian elimination**, named after Carl Friedrich Gauss. Numerical software packages, such as MATLAB, use elimination-based methods for many linear systems. Consider the following system of three linear equations:

$$\begin{aligned}x+2y+z&=2\\2x+6y+z&=7\\x+y+4z&=3\end{aligned} \tag{1.11}$$

We can replace the system by an equivalent matrix representation, which we call the **augmented matrix**.

$$M=(A|b)=\left(\begin{array}{cccc|c}a_{11}&a_{12}&\dots&a_{1n}&b_1\\a_{21}&a_{22}&\dots&a_{2n}&b_2\\\vdots&\vdots&\ddots&\vdots&\vdots\\a_{m1}&a_{m2}&\dots&a_{mn}&b_m\end{array}\right) \tag{1.12}$$

For example, the system

$$\begin{aligned}x+2y+z&=2,\\2x+6y+z&=7,\\x+y+4z&=3\end{aligned}\qquad\text{has augmented matrix}\qquad M=\left(\begin{array}{ccc|c}1&2&1&2\\2&6&1&7\\1&1&4&3\end{array}\right).$$

```
A = [1 2 1;2 6 1;1 1 4]
b = [2 7 3]'
M = [A b]
```

To eliminate M_{21}, take the 1st row, multiply it by 2, and then subtract the result from the 2nd row. These operations preserve the solution set because:

- Multiplying an equation by a nonzero scalar does not change its solution set.
- Adding a multiple of one equation to another preserves the solution set of the system.

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 2 & 6 & 1 & 7 \\ 1 & 1 & 4 & 3 \end{array}\right) -(\mathbf{2} \times R_1) \rightarrow \left(\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 0 & 2 & -1 & 3 \\ 1 & 1 & 4 & 3 \end{array}\right) \tag{1.13}$$

The number 1 in position $(1, 1)$ is called a **pivot**. We are relying on this number for the elimination process.

```
M(2,:) = M(2,:) - 2 * M(1,:)
```

Do the same for the 3rd row:

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 2 & 6 & 1 & 7 \\ 1 & 1 & 4 & 3 \end{array}\right) -(\mathbf{1} \times R_1) \rightarrow \left(\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 0 & 2 & -1 & 3 \\ 0 & -1 & 3 & 1 \end{array}\right) \tag{1.14}$$

```
M(3,:) = M(3,:) - M(1,:)
```

We continue by eliminating the second variable, y, from the third equation. To do this, we add half of the second equation to the third. The second pivot is the entry in position $(2, 2)$, which is 2. The result is

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 0 & 2 & -1 & 3 \\ 0 & -1 & 3 & 1 \end{array}\right) +(\mathbf{1/2} \times R_2) \rightarrow \left(\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 0 & 2 & -1 & 3 \\ 0 & 0 & 5/2 & 5/2 \end{array}\right) \tag{1.15}$$

```
M(3,:) = M(3,:) + .5 * M(2,:)
```

This gives the simpler system we seek. Its coefficient matrix is in **upper triangular form**, which we denote by U:

$$A \rightarrow U = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 2 & -1 \\ 0 & 0 & 5/2 \end{pmatrix} \tag{1.16}$$

An upper triangular system with nonzero diagonal entries can be solved by **back substitution**. Working upward from the last equation, we obtain

$$x = -3 \qquad y = 2 \qquad z = 1 \tag{1.17}$$

And that, barring a few minor complications that can crop up from time to time, is all that there is to the method of Gaussian elimination! It is extraordinarily simple, but its importance cannot be overemphasized.

1.3.1 When does elimination fail?

What if we encounter a 0 in the pivot position? For example:

$$A = \begin{pmatrix} 0 & 2 & 1 \\ 2 & 6 & 1 \\ 1 & 1 & 4 \end{pmatrix} \tag{1.18}$$

A zero entry cannot serve as a pivot. If a nonzero entry is available below it in the same column, we can exchange rows. Gaussian elimination with partial pivoting (GEPP) selects the entry with the largest absolute value among the remaining entries in that column and exchanges rows as needed. Now consider a different example:

$$A = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 1 & 4 \end{pmatrix} \tag{1.19}$$

Elimination produces a row of zeros, so no third pivot is available, even after row exchanges. We call such a matrix **singular** or **non-invertible**.

1.3.2 Elementary Matrices

Instead of manipulating the rows directly, as we did in the first example, we can represent each elimination step as a matrix multiplication. Let us focus on the coefficient matrix A. The first step in the example was:

$$\begin{pmatrix} 1 & 2 & 1 \\ 2 & 6 & 1 \\ 1 & 1 & 4 \end{pmatrix} -(\mathbf{2} \times R_1) \rightarrow \begin{pmatrix} 1 & 2 & 1 \\ 0 & 2 & -1 \\ 1 & 1 & 4 \end{pmatrix} \tag{1.20}$$

We ask which matrix, when multiplied by A on the left, produces the matrix on the right-hand side. In other words, we seek E_1 such that

$$E_1 \begin{pmatrix} 1 & 2 & 1 \\ 2 & 6 & 1 \\ 1 & 1 & 4 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 2 & -1 \\ 1 & 1 & 4 \end{pmatrix} \tag{1.21}$$

We now introduce a type of **elementary matrix**. Applying the row operation that adds -2 times the first row to the second row of the 3×3 identity matrix gives

$$E_1 = \begin{pmatrix} 1 & & \\ -2 & 1 & \\ & & 1 \end{pmatrix} \tag{1.22}$$

Consider the product row by row. The first row of E_1 must be $E_{1,1*} = (1, 0, 0)$ to leave the first row of A unchanged. The second row must be $E_{1,2*} = (-2, 1, 0)$ to subtract twice the first row of A from its second row. The third row must be $E_{1,3*} = (0, 0, 1)$ to leave the third row of A unchanged. Thus,

$$\begin{pmatrix} 1 & & \\ -2 & 1 & \\ & & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 1 \\ 2 & 6 & 1 \\ 1 & 1 & 4 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 2 & -1 \\ 1 & 1 & 4 \end{pmatrix} \tag{1.23}$$

Set

$$E_1 = \begin{pmatrix} 1 & & \\ -2 & 1 & \\ & & 1 \end{pmatrix} \qquad E_2 = \begin{pmatrix} 1 & & \\ & 1 & \\ -1 & & 1 \end{pmatrix} \qquad E_3 = \begin{pmatrix} 1 & & \\ & 1 & \\ & \frac{1}{2} & 1 \end{pmatrix} \tag{1.24}$$

Applying these elementary matrices in the correct order gives

$$A = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 6 & 1 \\ 1 & 1 & 4 \end{pmatrix} \quad \rightarrow \quad E_3E_2E_1A = U = \begin{pmatrix} 1 & 2 & 1 \\ & 2 & -1 \\ & & \frac{5}{2} \end{pmatrix} \tag{1.25}$$

```
E1 = eye(3); E1(2,1) = -2
E2 = eye(3); E2(3,1) = -1
E3 = eye(3); E3(3,2) = 1/2
U = E3*E2*E1*A
```

To undo the operation of adding c times row j to row i, we subtract c times row j from row i. The inverse elementary matrix has ones along the diagonal and $-c$ in position (i, j). Denote the inverses of the elementary matrices above by L_i:

$$L_1 = \begin{pmatrix} 1 & & \\ 2 & 1 & \\ & & 1 \end{pmatrix} \qquad L_2 = \begin{pmatrix} 1 & & \\ & 1 & \\ 1 & & 1 \end{pmatrix} \qquad L_3 = \begin{pmatrix} 1 & & \\ & 1 & \\ & -\frac{1}{2} & 1 \end{pmatrix} \tag{1.26}$$

```
L3 = eye(3); L3(3,2) = -1/2
L2 = eye(3); L2(3,1) = 1
L1 = eye(3); L1(2,1) = 2
L1 * L2 * L3 * U
```

Note that $L_1E_1 = L_2E_2 = L_3E_3 = I$. The product of the latter three elementary matrices is equal to

$$L = L_1L_2L_3 = \begin{pmatrix} 1 & & \\ 2 & 1 & \\ 1 & -\frac{1}{2} & 1 \end{pmatrix} \tag{1.27}$$

The matrix L is called a **lower unitriangular matrix**.

Lemma: If L and $\widehat{L}$ are lower triangular matrices of the same size, so is their product $L\widehat{L}$. The same holds for upper triangular matrices.

1.3.3 Permutations and Permutation Matrices

What about row exchanges? For instance, interchanging rows 1 and 2 of the 3×3 identity matrix produces the elementary interchange matrix

$$P = \begin{pmatrix} & 1 & \\ 1 & & \\ & & 1 \end{pmatrix} \tag{1.28}$$

```
P = eye(3);
P([1 2],:) = P([2 1],:)
```

The result PA of multiplying any 3-row matrix A on the left by P is the same as interchanging the first two rows of A. For instance,

$$\begin{pmatrix} & 1 & \\ 1 & & \\ & & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} = \begin{pmatrix} 4 & 5 & 6 \\ 1 & 2 & 3 \\ 7 & 8 & 9 \end{pmatrix} \tag{1.29}$$

Multiple row interchanges are accomplished by combining such elementary interchange matrices. The resulting row permutation is represented by a **permutation matrix**.

Examples of P:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \tag{1.30}$$

What is the inverse of a permutation matrix? How do we undo its row exchanges? For example, consider

$$\begin{pmatrix} & 1 & \\ 1 & & \\ & & 1 \end{pmatrix} \tag{1.31}$$

This matrix exchanges the first and second rows. Repeating the exchange reverses it, so the matrix is its own inverse. Now consider

$$\begin{pmatrix} & & 1 \\ 1 & & \\ & 1 & \end{pmatrix} \tag{1.32}$$

This matrix moves the original third row to the first position, the original first row to the second position, and the original second row to the third position. To undo this permutation, first exchange the second and third rows using

$$\begin{pmatrix} 1 & & \\ & & 1 \\ & 1 & \end{pmatrix} \tag{1.33}$$

and then exchange the first and third rows. The combined inverse permutation is

$$\begin{pmatrix} & 1 & \\ & & 1 \\ 1 & & \end{pmatrix} \tag{1.34}$$

This is the transpose of the original matrix. In fact, every permutation matrix satisfies $P^{-1} = P^T$.

1.3.4 The LU Factorization

We can write $L_1L_2L_3E_3E_2E_1A = A$. Since $L_1L_2L_3 = L$ and $E_3E_2E_1A = U$, we obtain

$$\begin{aligned} LU &= (L_1L_2L_3)(E_3E_2E_1A) \\ &= L_1L_2(L_3E_3)E_2E_1A = L_1L_2IE_2E_1A \\ &= L_1(L_2E_2)E_1A = L_1IE_1A \\ &= (L_1E_1)A = IA = A. \end{aligned} \tag{1.35}$$

In other words, we have factored the coefficient matrix as $A = LU$ into a product of a lower unitriangular matrix L and an upper triangular matrix U with the nonzero pivots on its main diagonal.

If a zero pivot requires a row exchange, we instead factor the permuted matrix:

$$PA = LU \tag{1.36}$$

So, it is PA that has an LU factorization.

Example: Find a permuted LU factorization of the matrix

$$A = \begin{pmatrix} 1 & 2 & -1 & 0 \\ 2 & 4 & -2 & -1 \\ -3 & -5 & 6 & 1 \\ -1 & 2 & 8 & -2 \end{pmatrix} \tag{1.37}$$

Eliminating the entries below the first pivot gives

$$\begin{pmatrix} 1 & 2 & -1 & 0 \\ 2 & 4 & -2 & -1 \\ -3 & -5 & 6 & 1 \\ -1 & 2 & 8 & -2 \end{pmatrix} \begin{matrix} \\ -2R_1 \\ +3R_1 \\ +R_1 \end{matrix} \to \begin{pmatrix} 1 & 2 & -1 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 1 & 3 & 1 \\ 0 & 4 & 7 & -2 \end{pmatrix} \qquad P = \begin{pmatrix} 1 & & & \\ & 1 & & \\ & & & 1 \\ & & 1 & \end{pmatrix} \qquad L = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ -3 & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{pmatrix} \tag{1.38}$$

We also need to keep track of P. We swap the 2nd and 3rd rows:

$$A = \begin{pmatrix} 1 & 2 & -1 & 0 \\ 0 & 1 & 3 & 1 \\ 0 & 0 & 0 & -1 \\ 0 & 4 & 7 & -2 \end{pmatrix} \qquad L = \begin{pmatrix} 1 & 0 & 0 & 0 \\ -3 & 1 & 0 & 0 \\ 2 & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{pmatrix} \qquad P = \begin{pmatrix} 1 & & & \\ & & 1 & \\ & 1 & & \\ & & & 1 \end{pmatrix} \tag{1.39}$$

We then eliminate the nonzero entry lying below the (2, 2) pivot, leading to

$$A = \begin{pmatrix} 1 & 2 & -1 & 0 \\ 0 & 1 & 3 & 1 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & -5 & -6 \end{pmatrix} \qquad L = \begin{pmatrix} 1 & 0 & 0 & 0 \\ -3 & 1 & 0 & 0 \\ 2 & 0 & 1 & 0 \\ -1 & 4 & 0 & 1 \end{pmatrix} \qquad P = \begin{pmatrix} 1 & & & \\ & & 1 & \\ & 1 & & \\ & & & 1 \end{pmatrix} \tag{1.40}$$

A final row interchange places the matrix in upper triangular form:

$$U = A = \begin{pmatrix} 1 & 2 & -1 & 0 \\ 0 & 1 & 3 & 1 \\ 0 & 0 & -5 & -6 \\ 0 & 0 & 0 & -1 \end{pmatrix} \qquad L = \begin{pmatrix} 1 & 0 & 0 & 0 \\ -3 & 1 & 0 & 0 \\ -1 & 4 & 1 & 0 \\ 2 & 0 & 0 & 1 \end{pmatrix} \qquad P = \begin{pmatrix} 1 & & & \\ & & 1 & \\ & & & 1 \\ & 1 & & \end{pmatrix} \tag{1.41}$$

Therefore,

$$PA = \begin{pmatrix} 1 & 2 & -1 & 0 \\ -3 & -5 & 6 & 1 \\ -1 & 2 & 8 & -2 \\ 2 & 4 & -2 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ -3 & 1 & 0 & 0 \\ -1 & 4 & 1 & 0 \\ 2 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & -1 & 0 \\ 0 & 1 & 3 & 1 \\ 0 & 0 & -5 & -6 \\ 0 & 0 & 0 & -1 \end{pmatrix} = LU \tag{1.42}$$

```
A = [1 2 -1 0;2 4 -2 -1;-3 -5 6 1;-1 2 8 -2]
L = eye(4)
P = eye(4)
U = A
L(2,1) = A(2,1)/A(1,1)
U(2,:) = U(2,:) - L(2,1)*U(1,:)
L(3,1) = A(3,1)/A(1,1)
U(3,:) = U(3,:) - L(3,1)*U(1,:)
L(4,1) = A(4,1)/A(1,1)
U(4,:) = U(4,:) - L(4,1)*U(1,:)
U([2 3],:) = U([3 2],:) % row exchange
P([2 3],:) = P([3 2],:) % keep track
L([2 3],1) = L([3 2],1)
L(3,2) = 0;
L(4,2) = U(4,2)/U(2,2)
U(4,:) = U(4,:) - L(4,2)*U(2,:)
U([3 4],:) = U([4 3],:)
P([3 4],:) = P([4 3],:)
L([3 4],[1 2]) = L([4 3],[1 2])
P*A - L*U
```

□

1.4 Computational Cost of GEPP

For a dense $N \times N$ matrix, elimination-based methods generally require $\mathcal{O}(N^3)$ arithmetic operations. Of course, in special cases, the actual arithmetic operation count might be

considerably reduced, particularly if A is a sparse matrix with many zero entries. A number of specialized techniques have been designed to handle sparse linear systems. A particularly important class consists of the tridiagonal matrices.

Figure 1.1 compares the measured time for a pivoted LU factorization of dense random $N \times N$ matrices with a reference curve cN^3, where N denotes the matrix size. The constant c is chosen so that the reference matches the largest-size measurement. Cubic growth means that doubling the matrix size requires roughly eight times as many arithmetic operations; on logarithmic axes, cN^3 is a straight line of slope 3. The timings approach the cubic reference for the larger matrices shown, while small problems are more affected by overhead and timing variability. These measurements illustrate the cost trend rather than an exact runtime law: hardware and implementation also affect elapsed time. The plot measures factorization only, not the subsequent triangular solves for a right-hand side.

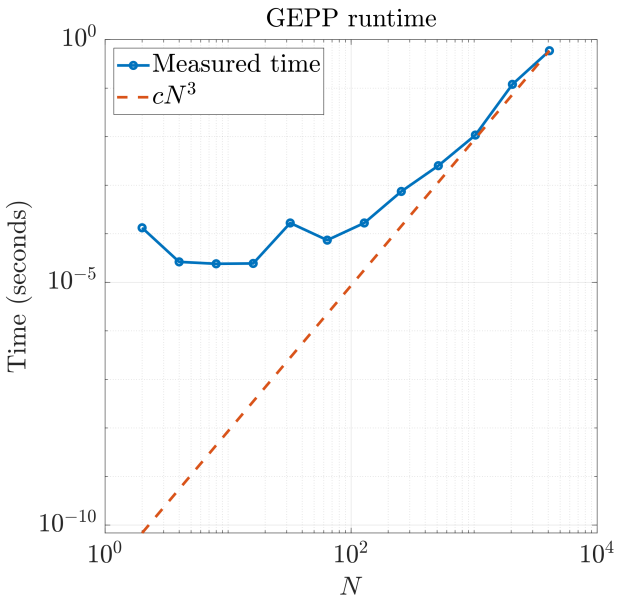


Figure 1.1: Measured pivoted LU factorization time and a cubic reference curve, matched at the largest matrix size.

1.5 Matrix Inverses

Definition: Let A be a square matrix of size $N \times N$. An $N \times N$ matrix X is called the inverse of A if it satisfies $XA = I = AX$. The inverse of A is denoted by A^{-1}.

Theorem: A square matrix has an inverse if and only if it is nonsingular.

Lemma: If A and B are invertible and of the same size, then AB is invertible and $(AB)^{-1} = B^{-1}A^{-1}$.

1.5.1 Gauss–Jordan Elimination

One algorithm for computing the inverse of a nonsingular matrix is **Gauss–Jordan elimination**. Instead of reducing A to triangular form, we reduce it to the identity matrix.

Example: Find the inverse of the matrix

$$A = \begin{pmatrix} 0 & 2 & 1 \\ 2 & 6 & 1 \\ 1 & 1 & 4 \end{pmatrix} \tag{1.43}$$

We form the augmented matrix

$$\left(\begin{array}{ccc|ccc} 0 & 2 & 1 & 1 & & \\ 2 & 6 & 1 & & 1 & \\ 1 & 1 & 4 & & & 1 \end{array}\right) \tag{1.44}$$

Applying Gaussian elimination gives

$$\left(\begin{array}{ccc|ccc} 2 & 6 & 1 & & 1 & \\ 0 & 2 & 1 & 1 & & \\ 0 & 0 & 9/2 & 1 & -1/2 & 1 \end{array}\right) \tag{1.45}$$

Up to this point, the procedure is the same as Gaussian elimination. We now divide each row by its pivot to make all pivot entries equal to 1:

$$\left(\begin{array}{ccc|ccc} 1 & 3 & 1/2 & & 1/2 & \\ 0 & 1 & 1/2 & 1/2 & & \\ 0 & 0 & 1 & 2/9 & -1/9 & 2/9 \end{array}\right) \tag{1.46}$$

We need to make the entries above the diagonal of the left-hand matrix 0. This can be done by elementary row operations, but now we work backwards. First, we eliminate the nonzero entries in the third column lying above the (3, 3) entry by subtracting $\frac{1}{2}$ times the third row from the second and also from the first:

$$\left(\begin{array}{ccc|ccc} 1 & 3 & 0 & -1/9 & 5/9 & -1/9 \\ 0 & 1 & 0 & 7/18 & 1/18 & -1/9 \\ 0 & 0 & 1 & 2/9 & -1/9 & 2/9 \end{array}\right) \tag{1.47}$$

Now use the second row to eliminate the entry in position $(1, 2)$:

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -23/18 & 7/18 & 2/9 \\ 0 & 1 & 0 & 7/18 & 1/18 & -1/9 \\ 0 & 0 & 1 & 2/9 & -1/9 & 2/9 \end{array}\right) \tag{1.48}$$

Therefore,

$$A^{-1} = \begin{pmatrix} -\frac{23}{18} & \frac{7}{18} & \frac{2}{9} \\ \frac{7}{18} & \frac{1}{18} & -\frac{1}{9} \\ \frac{2}{9} & -\frac{1}{9} & \frac{2}{9} \end{pmatrix} \tag{1.49}$$

□